

A “spaghetti sign” in feline abdominal radiographs predicts spleno-systemic collateral circulation

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Abstract

A well-defined, tortuous, and tubular soft tissue opaque structure has been observed within the left mid abdominal cavity in digital radiographs of some cats. Authors hypothesized that this radiographic “spaghetti sign” is a predictor for spleno-systemic collateral circulation. The purpose of this multicenter, retrospective, descriptive study was to further characterize radiographic “spaghetti signs” in a group of cats and compare these signs with results of ultrasonography or computed tomography (CT). A multiinstitutional database was reviewed for radiographic reports describing the presence of a tubular tortuous structure within the left abdominal cavity of feline patients. Patients were included if additional abdominal ultrasound or CT studies were available. Imaging studies and medical records were retrieved and reviewed by a board-certified veterinary radiologist, and characteristics were recorded. Eighteen cats met the inclusion criteria. The presence of a tortuous vessel consistent with spleno-systemic collateral circulation was observed by ultrasonography or CT examinations in all patients with a “spaghetti sign” on abdominal radiographs. All patients were spayed females. One cat showed ultrasonographic imaging features consistent with portal hypertension. In conclusion, findings supported the hypothesis that a radiographic “spaghetti sign” is a predictor of spleno-systemic collateral circulation in feline patients and should be distinguished from other signs such as focal loss of serosal detail or a space occupying lesion.

KEYWORDS

collateral circulation, feline, Radiography, spleno-systemic shunt

1 | INTRODUCTION

Digital radiography is now being used by most veterinary practices. This imaging modality provides increased contrast resolution compared to conventional radiology, accentuating the differences between tissue of different types and opacities.^{1,2} Image processing is used to reduce noise, remove artifacts, and further improve contrast resolution. Other advantages compared to conventional radiography are related to postprocessing image techniques such as spatial filtering and alteration in image contrast and brightness that further highlight differences between the opacity of tissues of different types.³ During digital radiographic examinations of some feline patients, authors have observed a tubular, tortuous, well-defined soft tissue structure within the left and mid abdominal cavity and have developed the term “spaghetti sign” for this finding. This structure is usually localized caudal to the spleen and lateral to the left kidney.

Spleno-systemic shunts have been recently described in feline patients, with spayed females being overrepresented.⁴ However, the

acquired or congenital origin of these shunts and their clinical significance has not been established. These vessels originate from the splenic vein and they terminate at the level of the left renal vein or caudal vena cava. Ultrasonographic and computed tomography (CT) characteristics of spleno-systemic shunts have been previously described in cats.⁴

Our hypothesis is that the soft tissue tubular tortuous structure (“spaghetti sign”), visible on digital radiographic examinations of the abdomen in cats, is a predictor of spleno-systemic collateral circulation. Objectives of the current study were to further characterize the radiographic “spaghetti sign” in a group of cats and compare this finding with results of ultrasonography or CT.

2 | MATERIALS AND METHODS

This is a multicentric retrospective descriptive study design. The clinical report databases of the Istituto Veterinario di Novara, Hope

Advanced Veterinary Center and the Clinica Veterinaria Castellarano were searched for feline patients of any age, sex, breed, and clinical history, from 2014 to 2016 having radiographic reports that described the presence of a tubular tortuous structure within the left abdominal cavity on the radiographic examination. Patients were included if concomitant abdominal ultrasound or CT studies were also available. Ultrasound and CT images were retrieved and retrospectively evaluated by a board-certified veterinary radiologist (S.S.). Presence of ultrasound and CT signs consistent with spleno-systemic collateral circulation were recorded for each cat and defined as presence of a tortuous vessel originating from the splenic vein with caudal direction and hepatofugal flow on Doppler evaluation.⁴ For cats with available Doppler examinations, presence of direct signs of portal hypertension (decreased flow velocity, less than 10 cm/s, in the portal vein with Doppler evaluation) were recorded.^{5–7} Indirect signs of portal hypertension were recorded from either ultrasound or CT studies and consisted of the following: thickening of submucosa of the gastric wall (gastric wall edema), presence of numerous hypoechoic stripes demarcating pancreatic lobulation and enlarged pancreas (pancreatic edema), ascites, gallbladder wall thickening with “double walled” appearance (gall bladder wall edema), subjective impression of increased dimensions of the left colic, cranial mesenteric, and gastroduodenal veins.^{7–14}

3 | RESULTS

Eighteen cats were included. All were female spayed domestic short-hair cats. The median age of the study group was 10 years (range 5–14 years) and the median body weight was 4.3 kg (range 3.5–5.2 kg).

Five cats were diagnosed with neoplasia affecting the gastrointestinal tract, the pancreas, maxillary bone, and mammary glands. Five cats were diagnosed with urinary tract diseases. A cytological evaluation of the liver was available in three patients, two with hepatic lipidosis and one with hepatic myelolipomas. One cat was diagnosed with gastrointestinal infection. A final diagnosis was not available in four cats. Detailed information about each case is reported in the Supplementary Files (Table 1).

The “spaghetti sign” was identified on all radiographic ventro-dorsal projections (18/18) (Fig. 1). The following radiographic machines were used: Pegasus DR (GE healthcare, Europe) set at 50–60 KV and 16–20 mAs (Istituto Veterinario di Novara), Cuattro DR (Heska, Switzerland) set at 80 KV and 2.5 mAs (Hope Advanced Veterinary Center) and FMI100-300 (Isomedic, Italy) set at 60 KV and 4 mAs (Clinica Veterinaria Castellarano). A lateral view was available in 15/18 cats, with 13/15 cats having a right lateral view, 1/15 cats having a left lateral view and 1/15 cats having both right and left lateral views. In four patients (4/18) the sign was visible on the right lateral projection caudal to the left kidney.

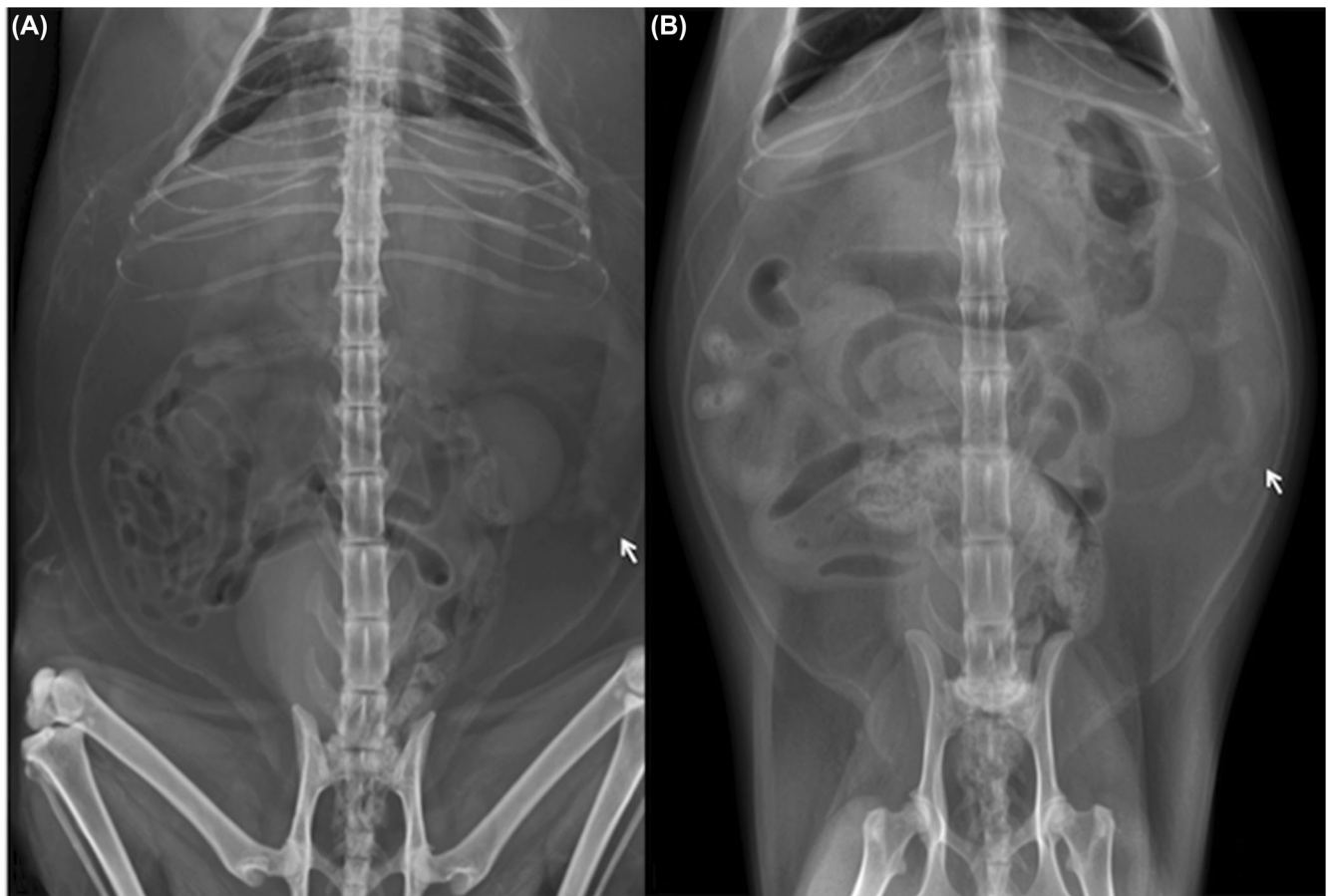


FIGURE 1 Ventro-dorsal radiographs of two different feline patients with visualization of the “spaghetti sign” localized lateral to the left kidney, caudal to the spleen, and extending toward the caudal abdomen (whites arrows in A and B)

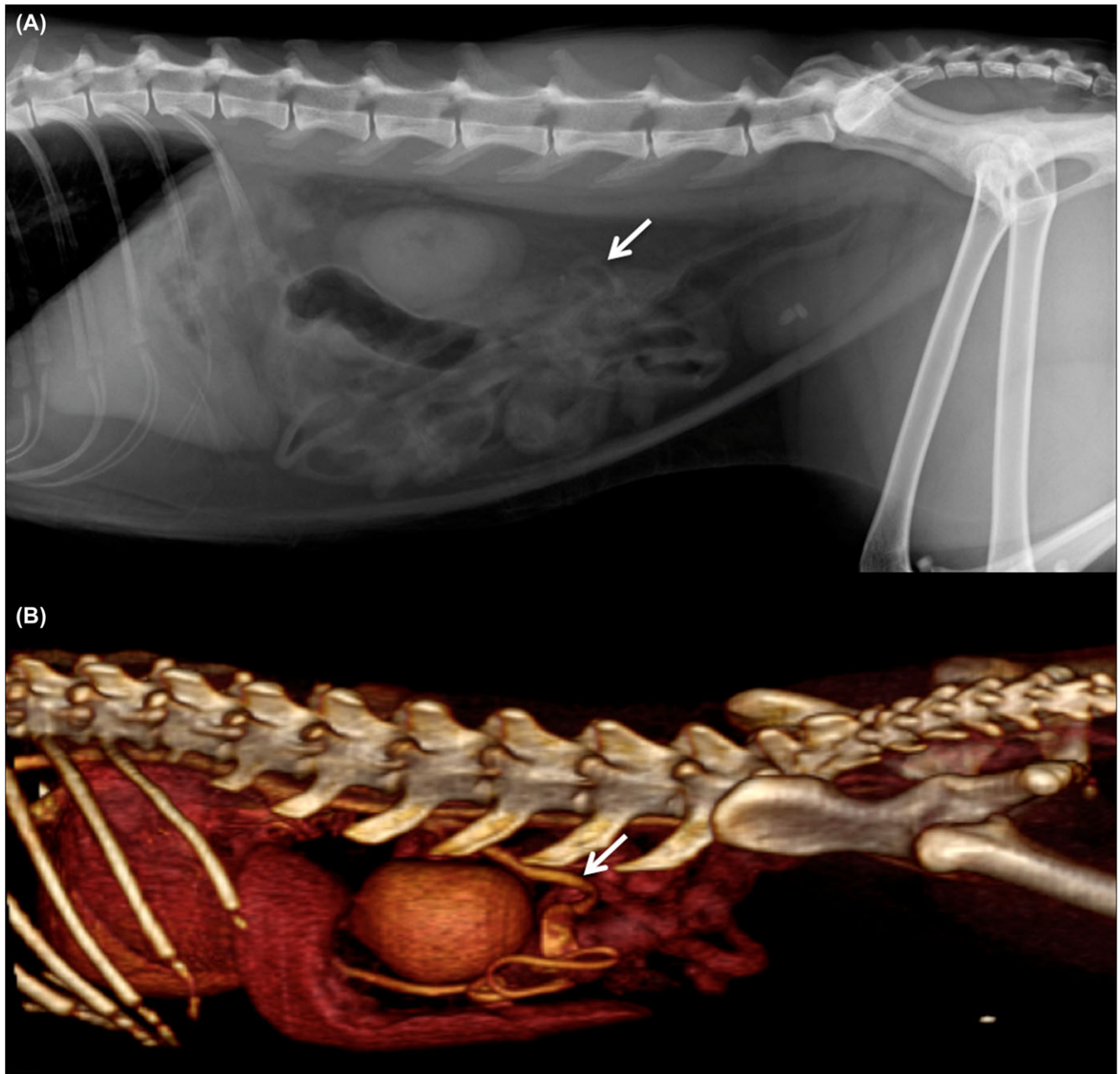


FIGURE 2 “Spaghetti sign” on the right lateral view (A) and corresponding 3D volume rendering computed tomographic image showing the presence of a spleno-systemic collateral circulation (white arrow) ending at the level of the left renal vein through the left gonadal vein (B) [Color figure can be viewed at wileyonlinelibrary.com]

Abdominal ultrasound was available in 18/18 patients and computed tomography in 2/18 patients. Ultrasonographic examination was performed with the following machines: Logiq S8 (GE healthcare, Europe) using a linear probe (8–18 MHz) (Istituto Veterinario di Novara), Logiq S8 (GE healthcare, Europe) using a linear probe (8–18 MHz) or a microconvex probe (8–11 MHz) (Hope Advanced Veterinary Center) and Toshiba Xario 200 using a linear (5–14 MHz), microconvex (4–10 MHz) probe (Clinica Veterinaria Castellarano). Computed tomographic images were obtained with a 16-slice helical MDCT scanner (Light Speed, GE, Italy). Postcontrast images were acquired after injection of 2 ml/Kg of iodinated non-ionic contrast medium (Omnipaque 350 mg/ml, GE Healthcare, Italy)

via a 22-gauge catheter into the cephalic vein, with flow rate at 2 ml/s, performed by a single barrel power injector system (Medrad, Italy).

Presence of a tortuous vessel originating from the caudal branch of the splenic vein consistent with spleno-systemic collateral circulation was observed with ultrasound or CT in all patients. This vessel ended at the level of the left renal vein through the gonadal vein in two patients with computed tomographic examination (Figs. 2 and 3). In cats where only ultrasound evaluation was available (16/18), it was not possible to recognize the termination of the spleno-systemic shunt due to the retrospective nature of the study vessel.

Doppler evaluation of the portal vein was available in two cats.

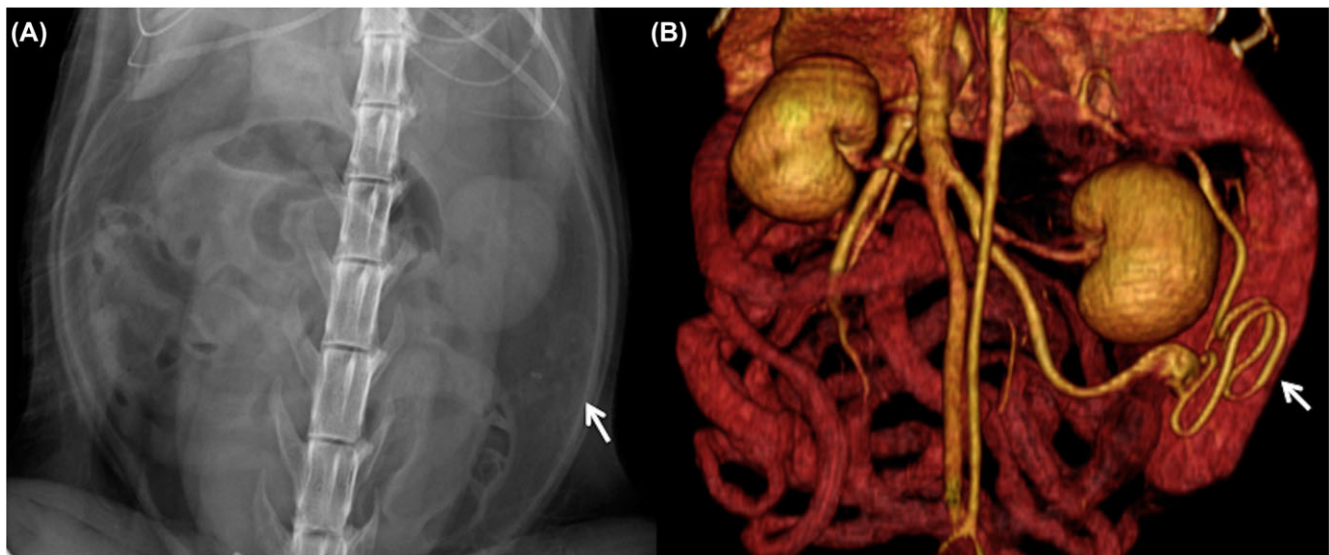


FIGURE 3 “Spaghetti sign” on the ventro-dorsal view (A) and corresponding 3D volume rendering computed tomographic image showing the presence of a spleno-systemic collateral circulation (white arrow) ending at the level of the left renal vein through the left gonadal vein (B) [Color figure can be viewed at wileyonlinelibrary.com]

One cat had thrombosis of the portal vein with secondary portal hypertension demonstrated by the presence of echogenic endoluminal material and decreased flow velocity (<10 cm/s) at spectral Doppler evaluation.^{5–7} One cat had spleno-systemic shunt with normal portal flow velocity at Doppler evaluation. The remaining patients did not show any clinical sign or indirect imaging features of portal hypertension.

4 | DISCUSSION

This study described the presence of ultrasound or CT findings consistent with spleno-systemic collateral circulation in a group of feline patients with a radiographic “spaghetti-sign” in the left mid to caudal abdominal cavity. In all cats, the “spaghetti-sign” was localized lateral to the left kidney, caudal to the spleen, and extended toward the caudal left abdomen. The left pancreatic limb is localized in the same anatomical area on the ventro-dorsal abdominal radiographs and the “spaghetti sign” should be distinguished from a focal loss of serosal detail or a space occupying lesion. This sign was seen on all the ventro-dorsal radiographic projections and in four of the eighteen lateral radiographic projections caudal to the kidneys in our sampled cats.

Several patterns of venous collateral circulation due to portal hypertension or caudal vena cava obstruction have been described in dogs and cats, in particular, a spleno-gonadic collateral circulation has been described with both hepatofugal and hepatopetal flow in canine patients with portal hypertension or caudal vena cava obstruction, respectively.¹⁴ In our cohort of cats only one patient showed portal vein thrombosis suggesting the spleno-systemic collateral circulation represented an acquired porto-systemic shunt. In the rest of the patients, no direct nor indirect imaging features or clinical signs of portal hypertension were detected. As previously reported, the spleno-systemic circulation was predominantly observed in female spayed

cats with no underlying portal hypertension.^{4,15} Authors of one study proposed that a spleno-systemic shunt in female spayed cats may be a consequence of adhesion formation from prior ovariohysterectomy.⁴

In our population of cats with spleno-systemic shunt, the shunting vessel drains blood from the spleen to the systemic venous circulation. This is different from patients with portal hypertension in which blood shunts in the systemic venous circulation also from gastrointestinal tract and other organs, and not only from the spleen. Although clinical signs of hepatic encephalopathy could not be completely ruled out in our cats, it is possible that this anatomical conformation of the shunt may induce less neurological signs compared to patients with portal hypertension and acquired portal collateral circulation.

Several limitations were inherent to our study. Due to the retrospective nature of the study, ultrasound, and CT images were not optimized for evaluation of spleno-systemic collateral circulation. In one previous study, 42% of cats had clinicopathological or histopathological changes that suggested hepatopathy with potential secondary portal hypertension.⁴ In our cohort of cats the cytological evaluation was available in only three cats of which one showed portal vein thrombosis and direct and indirect signs of portal hypertension. Due to the low amount of hepatic cytological evaluations we cannot definitely exclude the possibility of an underlying condition that may have created a previous transient portal hypertension. Furthermore, Doppler examination of the portal vein was available only in two cats.

In conclusion, findings from the current study supported the hypothesis that the “spaghetti sign” on ventro-dorsal, abdominal radiographs of feline patients is a predictor of ultrasound or CT characteristics of spleno-systemic collateral circulation. The “spaghetti sign” is localized within the anatomic region of the peripheral left pancreatic limb and should be distinguished from a focal loss of serosal detail or a space occupying lesion.

LIST OF AUTHOR CONTRIBUTIONS

Category 1

- (a) Conception and Design: Specchi S, Paek M
- (b) Acquisition of Data: Specchi S, Panopoulos I, Adrian AM, Spattini G, Morabito S, Paek M
- (c) Analysis and Interpretation of Data: Specchi S, Panopoulos I, Morabito S

Category 2

- (a) Drafting the Article: Specchi S, Panopoulos I, Morabito S
- (b) Revising Article for Intellectual Content: Specchi S, Panopoulos I, Adrian AM, Spattini G, Morabito S, Paek M

Category 3

- (a) Final Approval of the Completed Article: Specchi S, Panopoulos I, Adrian AM, Spattini G, Morabito S, Paek M

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SUPPORTING INFORMATION

Additional Supporting Information may be found online in the supporting information tab for this article.

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